

Lab Assignment 3: **Binomial Distribution**

Scenario:

Imagine a process where you flip a fair coin 10 times. Each flip is a Bernoulli trial (success = heads). Simulate the number of heads in each of 1000 such experiments.

Sub-questions:

(a) Simulate a Binomial Random Variable

1. Simulate 1000 experiments of flipping a fair coin 10 times.
2. Display the first 10 outcomes (i.e., number of heads per experiment).

(b) Frequency and Visualization

1. Count how many times each value (from 0 to 100) appears.
2. Count how many times each value (from 0 to 1000) appears.
3. Plot a bar chart of the frequency distribution.

(c) Estimating Probabilities

1. Compute the relative frequency for each value.
2. Compare these with the theoretical binomial probabilities.
3. Comment on how closely the simulation matches theory.